

The Sensation Modulating Network: an opponent architecture for embodied cognition, with haltability as the ground of object-directedness

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2 ABSTRACT

Cognitive science is divided between cognitivism, which explains the generativity of thought but detaches it from the body, and 4E approaches, which embody cognition but under-specify its mechanism and rarely commit to falsifiable predictions. We propose the **Sensation Modulating Network (SMN)**, an architecture in which cognition is the modulation of sensation through action. Its elementary unit is the *Sensation Modulator*, a tissue complex that senses and acts through one substrate; opponent pairs of these form *Coordinated Action Zones* that hold a dynamic equilibrium against gravity and the medium. We give the single-zone dynamics as a closed formal system, including a *haltability* operator whose state variable — *alert energy* — is the real metabolic cost of holding an antagonistic configuration. We argue that haltability is the architectural ground of *object-directedness*: an object is registered as *other* when an unpredicted resistance halts an ongoing action. Reafference separates self from world; a broadcast operator coordinates a decentralized network; and layered action patterns — basal, halttable, and negotiable — connect the grounded body to the recombinable structure the cognitivist tradition prizes, the last stated as a hypothesis whose demonstration we defer to a companion paper. The architecture yields eight measurable *registers* (architectural fingerprints), and we report a confirm-and-correct ledger from an open simulation bench, SMN-Lab, that tests them. We position the SMN by its affinities and differences with active inference, Global Workspace Theory, Gärdenfors's conceptual spaces, and Levin's basal cognition, and we name the wet-lab measurement that would decide its central energetic claim.

Keywords: embodied cognition, opponent control, reafference, haltability, morphological computation, action patterns, conceptual spaces, falsifiability

1 INTRODUCTION

Two traditions divide the science of mind. **Cognitivism** explains the generativity of thought — its recursive, compositional productivity — but treats the body as peripheral input and output to a central computation. **4E cognition** (embodied, embedded, enactive, extended) returns thought to the body and its world, but has been weaker on the nervous system's specific organization and, more consequentially for a science,

weaker on *falsifiable mechanism*. Neither tradition has produced a unified account that explains both the grounded character of animal cognition and the generative character of human thought, and that commits to predictions it could fail.

This paper offers such an account and states it so that it can be tested. The proposal is the **Sensation Modulating Network (SMN)**: an architecture in which cognition is the *modulation of sensation through action*. The architecture is built from one repeated element — a tissue complex that senses and acts through a single substrate — organized by opponency into a body-spanning network that holds itself in dynamic equilibrium against the fields it lives in. We summarize the architecture compactly, give its single-zone dynamics as a closed formal system, and derive measurable signatures from it; we then report computational tests and locate the account among its neighbours.

The organizing thesis concerns **haltability**. Classical accounts of action emphasize initiation; the SMN emphasizes the active *hold*. A body built from opponent pairs can recruit a co-active equilibrium in which two pulls cancel and a configuration is held still — not by relaxing and not by inhibiting one side, but by spending energy to maintain a balance. We call this capacity haltability and its state variable *alert energy*, a non-negative, dimensionful quantity that measures the metabolic cost of the hold. The central claim is that haltability is the architectural ground of *object-directedness*: when an ongoing action meets a resistance it did not predict, the same antagonistic substrate that can hold a posture halts the action, and in that halt a portion of the world is registered, for the first time, as other than the self that was acting. Aboutness does not require a symbol pointing at an object; it requires a body whose action can be stopped by what it is about.

The paper makes three contributions. First (Sections 2–5), it specifies the architecture and brings its formalism into the open: the Sensation Modulator, the Coordinated Action Zone with its haltability operator, reafference, the broadcast network, and the layered action patterns. Second (Section 6), it derives eight *registers* — concrete, measurable behavioural signatures that the architecture either exhibits or does not. Third (Sections 7–9), it positions the account by its affinities and differences with neighbouring frameworks and reports a confirm-and-correct ledger from an open bench (SMN-Lab) that puts the registers at risk.

A note on scope. The architecture reaches from basal autonomic patterns up to the negotiable, recombinable patterns we argue ground generative structure; but the full account of that last step — how negotiable action patterns underwrite generative grammar — is the subject of a companion paper (Nagarjuna and Karnam, 2026) and is here stated only as a hypothesis (Section 5). The architecture brackets phenomenology throughout: it specifies the structural conditions object-directedness requires, not the felt character that may occupy them.

2 THE ARCHITECTURE BY COMMITMENTS

The SMN rests on six commitments, of two kinds: empirical generalizations the architecture takes on, and structural or design commitments about its elements.

- **C1 (opponency at every scale).** The body is organized by opponent dynamics at every scale at which it is organized — biochemical, autonomic, motor, bilateral/segmental, neural, perceptual. The architectural primitive at every scale is an opponent pair whose dynamics produce a balance state. This is an empirical observation, not a modeling convenience.
- **C2 (Sensation Modulators everywhere).** The smallest element is the Sensation Modulator (SM): a single tissue complex that both senses and acts through one substrate, with a mechanical interface

engaging body geometry and a messaging interface engaging the network. The architecture is neutral about what carries the messages but commits to the structural fact that sensing and acting share a substrate.

- **C3 (decentralized coordination through broadcast).** Each CAZ runs its own balance computation locally, from its own afferents and the broadcast it reads from the rest of the body. Coordination is a property of the network, not a function any centre computes.
- **C4 (active-perception filtering).** Each board admits to its predictive computation only sensor streams currently coupled to its modulation; uncoupled streams remain under low-cost peripheral monitoring. With a body producing far more data than local computation can attend to, the architecture filters at ingress (O'Regan and Noë, 2001).
- **C5 (habituation).** Streams that are coupled but fully predicted by an ongoing basal pattern — heartbeat, breathing, postural sway — drop out of attended status because their coupling carries no residual information.
- **C6 (spatial cognition through differential displacement).** Spatial structure is constructible only from the differential displacement the modulators produce, registered against the axes the body's developmental asymmetries establish. The SMN is fundamentally geometric: what it constructs is a body-anchored spatial model of itself and the world.

C1 and C6 are empirical generalizations that sustained contrary evidence would falsify; C2 and C3 fix the elementary units and their coherence; C4 and C5 are parsimony commitments made at instantiation. The architecture is composed by repetition: opponent pairs of Sensation Modulators form Coordinated Action Zones, and a broadcast integrates many zones into one network (Figure 1). Control is *morphological computation* (Pfeifer and Bongard, 2006): the coupled body performs work a central controller would otherwise do.

3 THE SENSATION MODULATOR AND THE COORDINATED ACTION ZONE

3.1 The Sensation Modulator

The Sensation Modulator is specified by two coupled relations:

$$F = \Phi(a, q, \dot{q}), \quad F \geq 0 \text{ (pull-only)} \quad (1)$$

$$s = \Psi(q, \dot{q}, F). \quad (2)$$

Equation (1) states that the force the element produces depends jointly on the activation it receives and its configuration and velocity, with the pull-only constraint reflecting the biological fact that muscle and analogous tissues pull but do not push. Equation (2) states that the afferent the element emits is read out by receptors embedded in the *same* tissue whose deformation produces F — so the same q that sets F via Φ also sets s via Ψ . There is no internal boundary between “the sensor” and “the actuator”; this is the *dual interface*, supported by the co-placement of muscle spindles, Golgi tendon organs, and joint receptors (Proske and Gandevia, 2012). The commitments are about the *structure* of Φ and Ψ , not their parametric form: any (Φ, Ψ) that is pull-only and reports a readable configuration variable is an admissible realization (vertebrate or invertebrate muscle, or artificial actuators such as pneumatic or dielectric-elastomer “muscles”), and the architecture's downstream claims survive substitution.

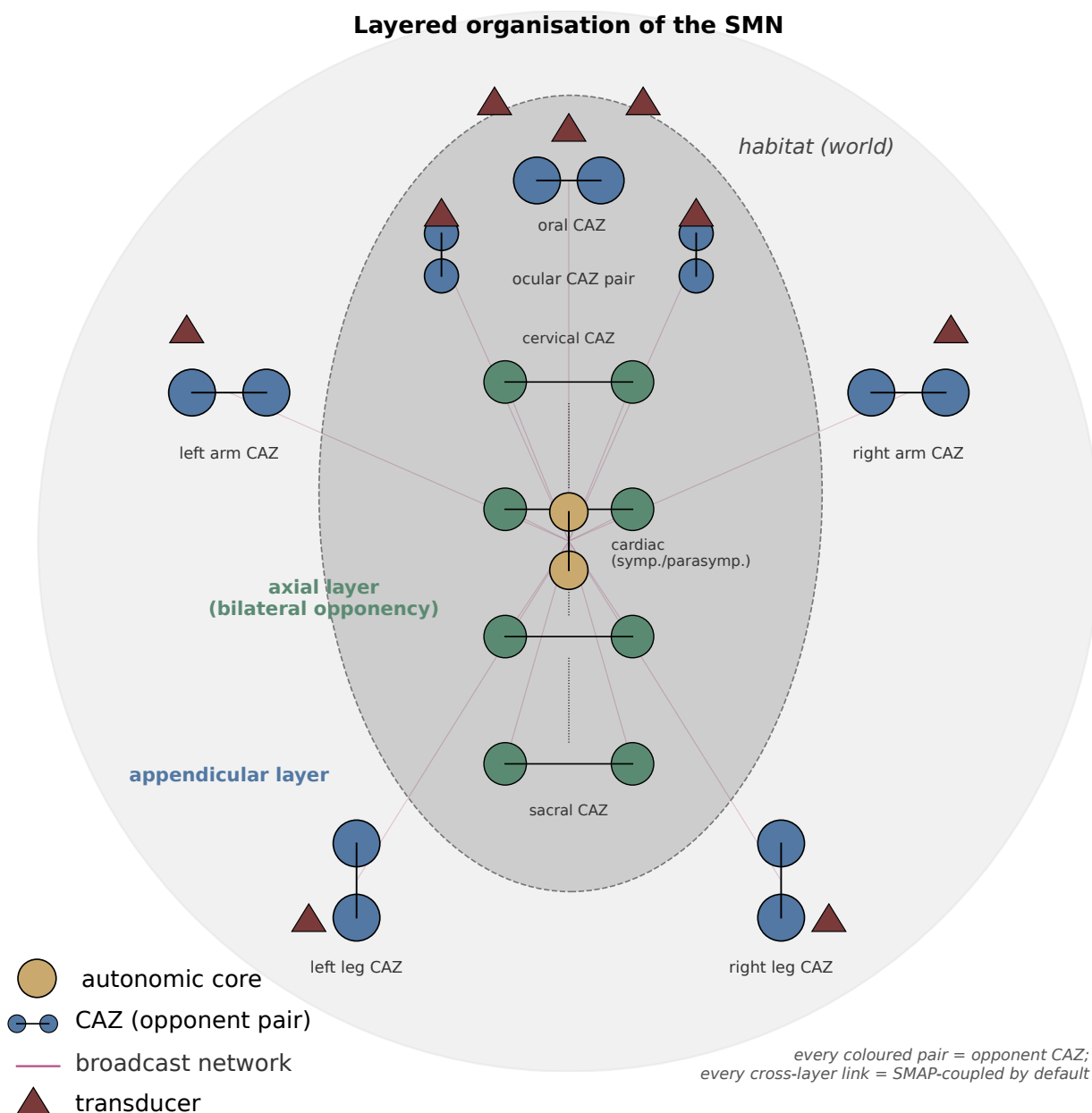


Figure 1. The SMN at three nested scales. The Sensation Modulator senses and acts through one substrate (a mechanical interface engaging body geometry, a messaging interface engaging the network). Two opponent modulators and a communication board form a Coordinated Action Zone that holds a joint in dynamic equilibrium. Many zones, integrated by a body-wide broadcast, form the network; each level is composed from the previous by opponent pairing and broadcast.

103 3.2 The Coordinated Action Zone

104 A Coordinated Action Zone (CAZ) is the tuple (Z_+, Z_-, B, H) : two opponent modulators, a
105 communication board B , and a haltability operator H , sharing a configuration $(q, \dot{q})$ and an internal
106 alert energy E_R . For a one-degree-of-freedom joint with moment arm r and inertia I ,

$$I \ddot{q} = r (F_+ - F_-) - b \dot{q}, \quad (3)$$

with $F_{\pm} = \Phi(a_{\pm}, \ell_{\pm}, \dot{\ell}_{\pm})$ and $\ell_{\pm} = \ell_0 \mp rq$. What is non-standard is that a_+ and a_- are not independent commands from a controller: they are emitted by the board. The agent is placed in a gravitational field treated as a *structural prior* — an ambient ordering the body did not choose — and the opponent pair is organized to oppose it, holding a *dynamic equilibrium* that action departs from rather than a setpoint a controller tracks.

3.3 The communication board

The board does three things: it *reads* the afferent each zone emits, it *routes* that afferent to the partner zone (the inverse coupling that makes the two zones behave like the pans of a balance), and it *applies modulation* — a task drive, a halt gate, and an alert-energy bias. With $g \in \{0, 1\}$ the halt gate ($g = 1$ active, $g = 0$ halted), λ the task equilibrium, and $a_0 > 0$ the baseline co-contraction,

$$g = 0 : a_+ = a_- = a_0; \quad g = 1 : \text{drive} = K_p(\lambda - q) - K_d\dot{q}, \quad (4)$$

with the active and partner activations set by the drive sign and biased by E_R through coefficients (β, γ) (full routing in the Appendix). The board is not a message bus carrying addressed packets, and the broadcast it reads from is not a non-physical signal from a centre. The broadcast is *physically realized but functionally individuated and medium-independent*: every message is a physical state — a tension, a rate, a potential — in a shared medium that any zone writes to and reads from by changing a shared variable (a hormonal milieu, a bioelectric field, global postural tension). A signal here is just a difference in one part of the body that makes a difference to another; nothing contentful is shipped and nothing reads it from outside.

3.4 Haltability and alert energy

Haltability is the capacity to actively hold a configuration still, distinct from passive relaxation and from classical inhibition. It is a property of the antagonism between the two zones, not something the board generates: two pull-only zones whose forces oppose along a shared variable *admit* a co-active equilibrium in which the forces cancel and the configuration is held without motion — as the coupling of a constrained mass makes a pendulum's stable attractor exist. The board does not produce that state; it *recruits* the affordance and maintains it. The state variable of the recruitment is the *alert energy* $E_R \geq 0$:

$$\frac{dE_R}{dt} = \rho F_{\text{active}} \mathbf{1}[g = 1] - \frac{E_R}{\tau_E}, \quad (5)$$

where $F_{\text{active}} = \max(F_+, F_-)$, ρ is the build rate, and τ_E the decay constant. Alert energy accumulates while a zone is driven, in proportion to its load, and decays passively otherwise; its loaded steady state is

$$E_R^* = \rho \tau_E \langle F_{\text{active}} \rangle. \quad (6)$$

Three consequences follow. *Alert energy is a real metabolic quantity*, not a free parameter: bounded, dimensionful, passively decaying, and predicted to be measurable as the metabolic difference between an engaged-but-not-active zone and the same zone at rest. *Active equilibrium* (the inactive zone tonically engaged in proportion to the active zone's load) is the loaded steady state of (5). And the *distinction from classical inhibition is sharp*: classical inhibition has no alert energy — the inactive zone is silenced, with no recruited affordance to maintain. The dynamics of E_R are what the SMN claims biological tissue does and reciprocal inhibition does not, and they yield the load-coupling and post-release signatures of Section 6.

4 TRANSDUCER, REAFFERENCE, AND DIRECTEDNESS

4.1 The transducer and the refference principle

Unlike the Sensation Modulator, a *transducer* is read-only: it maps a physical field to a signal, $S : \Phi(t, x) \mapsto s(t)$, without writing back to the body. For an agent with configuration θ and a stimulus at φ , a Gaussian receptive field gives

$$s(t) = I \exp\left(-\frac{(\theta(t) - \varphi(t))^2}{\sigma^2}\right) + \eta(t), \quad (7)$$

with s depending jointly on θ (the agent's configuration, modulable by action) and φ (the world, only conditionally observable). A transducer can register that something is happening; it cannot, alone, say whether the agent or the world caused it. The SMN resolves this with the *refference principle* (von Holst and Mittelstaedt, 1950; Crapse and Sommer, 2008): a predictor in the board uses the directly observed θ and the agent's world estimate $\hat{\varphi}$ to predict $\hat{s} = I \exp(-(\theta - \hat{\varphi})^2/\sigma^2)$ and forms the *refference residual*

$$r(t) = s(t) - \hat{s}(t). \quad (8)$$

The residual is small for self-generated change (the predictor tracks) and large for world-generated change (the estimate is outdated); it is the architecture's primitive signal of *self versus world*. When the world estimate adapts by a gradient step $\dot{\hat{\varphi}} = \kappa r / (\partial \hat{s} / \partial \varphi)$, the post-change residual decays — and that decay is mastery of a sensorimotor contingency in O'Regan and Noë's sense (O'Regan and Noë, 2001), now a measurable adaptation rather than a stipulated competence.

4.2 Surprise as attention; halt as directedness

The residual also feeds alert energy, $dE_R/dt = \rho F_{\text{active}} \mathbf{1}[g=1] + \rho_r |r| - E_R/\tau_E$, and an unpredicted event auto-halts the drive when $|r| > \tau_h$. The *same* alert-energy substrate that distinguishes haltability from inhibition is thus the substrate of attention. This yields an interruption story — surprise gates action — but the thesis claims more: that haltability is the condition object-directedness requires. The recruited halt is not stillness; it is a configuration *held available* (a stance toward a state, not a generator of one), *held interruptibly* (releasable by the world, unlike a thermostat, which cannot be surprised), and *held under prediction* (the residual reports that *this* thing is deviating from expectation). Together these are *architectural directedness* — the structural form of being directed at something. This is the architectural form of what Smith calls *registration*: objecthood is an achievement of engaged participation, not a category the world supplies and the mind detects (Smith, 1996).

4.3 An embodied two-vector model

The account has a close affinity with Gärdenfors's analysis of events in conceptual spaces (Gärdenfors, 2020, 2019). An SMN agent pressing on the world is an embodied realization of the two-vector model: the antagonist pull is the *force vector*, the object's response (it yields, resists, or complies) is the *result vector*, and gravity and friction enter as counterforces. Objecthood, for this agent, simply *is* the force-to-result mapping, sensed as resistance to its own modulation. The one refinement we add is that the agent does not *learn* the mapping probabilistically: it *enacts* it by construction, through the opposition of its pulls against the field. The force is physically real; what is constructed is the objecthood the agent reads off it.

173 4.4 What sets the focus, and the status of the world estimate

174 Two clarifications. First, haltability *resolves* a focus but does not *originate* the caring that makes a
 175 focus worth having; what sets which configuration is worth holding comes from the organism's biological
 176 normativity (what matters to a self-maintaining body) and its sedimented experiential history, not from
 177 the halt mechanism itself. Second, the world estimate $\hat{\varphi}$ is meant deflationary: for the living agent there
 178 is no stored, contentful world-model, only a current attunement constituted in coupling. "Model" in the
 179 representational sense is reserved for the artificial case, where structure genuinely is stored and positional
 180 — an asymmetry between natural and artificial cognition we develop elsewhere.

5 NETWORKS, BROADCAST, AND THE LAYERED ACTION PATTERNS

181 5.1 The dual-signal property

182 A multi-zone body runs coordinated activation trajectories. Write $y_t = F(\theta_t; z_t)$ for the body's own
 183 modulation map from its joint configuration θ_t to its afferent vector, with z_t the fixed background. A
 184 trajectory is *dual-signal* when two zones Z_x, Z_y are sensitive to a common parameter θ (DS-1), both
 185 efference copies are available to the predictor through the broadcast (DS-2), and the pattern is agent-
 186 halttable (DS-3). The dual-signal property makes self-contact, world-contact, and contact with another agent
 187 into three *structurally distinct* residual signatures, distinguishable by the form of the residual rather than its
 188 magnitude — the architectural basis of the self/world/other distinction (registers 5 and 8, Section 6).

189 5.2 The phase-oscillator picture and four regimes

190 A chain of N segmental zones is described by Kuramoto-style coupled oscillators with modulatory input
 191 (Kuramoto, 1984):

$$\dot{\theta}_i = \omega_i + \sum_{j \in \mathcal{N}(i)} K_{ij} \sin(\theta_j - \theta_i - \varphi_{ij}) + u_i(t), \quad (9)$$

192 where u_i is not stipulated but computed by the i th zone's board and haltability operator. Four regimes
 193 follow: a **Basal Action Pattern** (BAP; $u_i \equiv 0$, a stable limit cycle — heartbeat, gait, breathing); a **Halttable**
 194 **Action Pattern** (HAP; u_i a halt pulse, with E_R governed by (5)); a **Negotiable Action Pattern** (NAP; u_i
 195 continuously rephases or reweights the coupling in response to perceptual predicates); and a **Transactional**
 196 **Action Pattern** (TAP; a NAP whose halt events are externalized as a public trace recoverable by other
 197 agents).

198 5.3 The network and the broadcast

199 The full network is the tuple $\text{SMN} = (G, \mathcal{Z}, \mathcal{B}, \mathcal{H}, \mathcal{S}, \Pi, \alpha)$: a routing graph G over zone nodes $\mathcal{Z}$, the
 200 local boards $\mathcal{B}$ and haltability operators $\mathcal{H}$, the sensor space $\mathcal{S}$, the *broadcast operator* Π that makes
 201 one zone's state available to another's predictor, and the active-perception filter α of C4. The broadcast
 202 is the decentralized substitute for a central controller (C3): it maintains a shared state-space that every
 203 zone contributes to and reads from. The single-zone dynamics above are specified as a closed system;
 204 the analytical theory of multi-zone networks — stability, controllability, convergence — is, by our own
 205 statement, deferred, and the computational bench of Section 8 is where the network-level claims are
 206 exercised in the interim.

5.4 From negotiable patterns to generative structure (hypothesis)

The basal and haltable patterns above are fully within the present account. The negotiable patterns point beyond it. NAPS are modifiable, recombinable primitives: a zone or sub-network can rephase, regate, and re-route on the fly, and nested zones can be recruited and released as units. We hypothesize that this recombining ability — CAZ switching over a nested architecture, with externalized traces (TAPS) crossing body boundaries — is where the recursive, generative structure the cognitivist tradition locates in grammar (Chomsky, 1957) becomes architecturally tractable: generativity grounded in body-anchored modular dynamics rather than in disembodied symbols. We state this as a hypothesis and defer its demonstration to a companion paper (Nagarjuna and Karnam, 2026); the present paper establishes the BAP and HAP layers it would build on.

6 THE EIGHT PREDICTED REGISTERS

The architecture commits to eight *registers* — measurable signatures, each derived from the formalism, that a real or simulated system either exhibits or does not. They are the architecture's falsifiers.

1. **Tonic load coupling.** The inactive zone's tonic activation rises linearly with the active zone's load, $a_{\text{partner}} \approx a_0 + \beta \rho \tau_E F_{\text{active}}$, where classical inhibition predicts independence of load.
2. **Resumption latency.** After a brief halt, latency to a small reversed target is shorter than under classical inhibition by $\Delta t \approx \gamma E_R(t_{\text{release}})/K_p$.
3. **Reafference (self vs world).** For a fixed sensor change the residual is small when self-generated and large when world-generated; in the noiseless limit the self-residual is zero.
4. **Sensorimotor-contingency mastery.** With world-model adaptation on, the post-change residual decays to the noise floor; the decay rate is mastery, made measurable.
5. **Intra-body dual-signal residual.** Regressing the residual on the partner-as-stimulus form gives $R^2 \rightarrow 1$ for two zones of one SMN sharing a substrate, and $R^2 \rightarrow 0$ for an exogenous stimulus — a structural, not magnitude, signature.
6. **Surprise-driven attention and auto-halt.** An unpredicted event drives the residual above threshold and gates the drive within one step; attention and haltability share one substrate.
7. **Shared intentionality (hypothesis).** Two agents in mutual reafference, both adapting, show a joint residual that decays far faster than under asymmetric or no adaptation (Tomasello et al., 2005).
8. **Self / world / other.** The three dual-signal indices give a three-way structural distinction between broadcast-internal contact, contact with a non-intentional exterior, and contact with another agent's structured modulation.

Three further consequences of the network formalism (basal-pattern entrainment of haltable zones, axial-to-appendicular mode-switching, Hebbian recruitment of negotiable patterns) presuppose a multi-zone implementation and are flagged as targets rather than numbered registers.

7 AFFINITIES AND DIFFERENCES

The SMN shares commitments with several well-developed programs, and locating it among them is best done by naming what it shares and where it diverges, not by displacing anything.

242 7.0.0.1 Active inference and predictive processing.

243 The relationship is more reconcilable than an oppositional reading suggests (Friston, 2010; Parr et al.,
 244 2022). The mathematics is largely shared: the refference residual $r = s - \hat{s}$ is structurally a prediction
 245 error, equation (5) couples surprise to attention and auto-halt, and registers 3 and 6 are contrasts active
 246 inference also predicts. Both accounts are hierarchical, and both make the body act as a unit — active
 247 inference through a global free-energy quantity, the SMN through the broadcast operator Π that integrates
 248 local zones into a shared state-space. Indeed the activated subset of the network (the zones currently
 249 coupled through haltability and broadcast for a task) can be read as a Markov blanket, so that the two are
 250 plausibly *dual descriptions of one object*. The differences appear to turn on emphasis rather than opposition:
 251 the SMN is *constructive* (the architecture is specified first and prediction-error dynamics fall out) where
 252 active inference is in part *normative* (agents minimize free energy and the architecture is whatever does
 253 so), and the SMN is decentralized, with no single quantity the network optimizes. The one place where
 254 the SMN ventures a commitment its neighbour need not share is alert energy: a *real metabolic cost* of the
 255 hold, with the load-coupling and post-release dynamics of registers 1 and 2 — a positive, measurable claim
 256 (Section 9).

257 7.0.0.2 Global Workspace Theory.

258 The broadcast operator Π gives a concrete structural mechanism for the kind of shared access Global
 259 Workspace Theory describes (Baars, 2005; Dehaene, 2014). The two are pitched at different levels —
 260 the SMN at the architecture of coordination, Global Workspace at access — so the contribution here is
 261 to ground “broadcast” in a network operation with measurable consequences (the dual-signal indices of
 262 register 8) rather than in a workspace metaphor.

263 7.0.0.3 Opponent and equilibrium-point control.

264 The closest kin in motor control is the equilibrium-point tradition (Feldman, 1986; Latash, 2008):
 265 movement as a shift of the equilibrium of antagonist pairs rather than a computed torque. The SMN
 266 inherits this and adds the costly, actively maintained hold — haltability and alert energy — that the
 267 equilibrium-point account does not price.

268 7.0.0.4 Conceptual spaces (Gärdenfors).

269 As developed in Section 4, the SMN agent is an embodied realization of the two-vector model, and the
 270 body-relative state-space it constructs is, in Gärdenfors’s terms, a conceptual space (Gärdenfors, 2020,
 271 2019). We read conceptual spaces as the bridge from *grounded* to *positional* meaning — the level at which
 272 embodied structure becomes a navigable geometry — a thread we develop in the companion work.

273 7.0.0.5 Basal cognition (Levin), an ally.

274 The SMN’s claim that the broadcast is physically realized but medium-independent (Section 3) has a
 275 natural realizer in the endogenous bioelectric networks Levin and colleagues describe, where non-neural
 276 tissue stores, reads, and acts on shared electrical state (Levin, 2014; Levin and Pezzulo, 2016). Basal
 277 cognition is an ally: cognition without a brain, organized by opponent and field dynamics, is exactly what
 278 the SMN’s commitments predict should be possible.

279 7.0.0.6 Enactivism and registration.

280 Finally, the SMN is a friend of sensorimotor enactivism, supplying a mechanism (register 4) for perception
 281 as the mastery of sensorimotor contingencies (O’Regan and Noë, 2001), and it cashes out Smith’s account

of objecthood as *registration* through engaged participation (Smith, 1996) in the dual-signal property of Section 5.

8 COMPUTATIONAL TESTS: THE SMN-LAB LEDGER

The architecture's single-zone dynamics are specified; its multi-zone theory is deferred. To exercise the network claims in the interim — and to put the registers at risk — we built **SMN-Lab**, an open simulation bench on the MuJoCo physics engine, organized around a *derived* minimal model organism (the smallest body that can initiate non-inertial movement). Each study is pre-registered with a matched non-modulatory foil (identical except for the one thing the architecture says matters), replicated across seeds; adjustments beyond the published architecture are declared, and confounded runs are discarded rather than reported. Full methods, parameters, and seeded datasets are at <https://smn-lab.readthedocs.io>; here we report only the ledger (Table 1).

Table 1. The confirm-and-correct ledger across nine pre-registered SMN-Lab studies; three further confounded runs were caught and discarded before reporting.

Study	SMN prediction	Outcome
S0 coupling	locomotion is a network effect	confirmed (across-seed variance collapses $\sim 100\times$)
S1 geometry $\rightarrow$ world-model	body-relative model, scales with geometry	model confirmed; scaling <i>corrected</i> (flat in segment count)
Q1 modulation	modulation carries self/world resolution	core confirmed; one sub-claim <i>corrected</i>
Q1b resolution	resolution scales with zone density, with modulation	confirmed ($\sim 14 \rightarrow 22.6$, $\propto 1/\sqrt{N}$)
Q2 reafference	self/world separation by reafference	partial (37% cancellation; distributed-body limit)
Pred-1 haltability	a distinct active-halt signature	confirmed (dwell ~ 0.17 s)
Pred-2 zonal	substrate-specific priors required	partial (strong reading not supported)
Pred-3 antagonism	antagonistic benefit at a load-scaled cost	confirmed ($-28\times$ integrated error; $\sim$ quadratic cost)

The arithmetic is the point: across nine pre-registered studies the architecture was confirmed about as often as it was corrected. The two corrections that bite — the world model does *not* improve with raw geometry (S1), and self/world resolution *collapses* without modulation (Q1) — converge on a single sharpened claim, the *resolution principle*: the architecture's explanatory work is done by modulation, not by transduction. An instrument that only ever confirmed its theory would be an illustration; this one is a test.

9 DISCUSSION

9.0.0.1 What the bench buys, and the gaps it answers.

The multi-zone studies above are working demonstrations of the network the analytic theory leaves open (S0, Q1/Q1b, and the antagonistic-control results all exercise coupling, density, and opponent dynamics across many zones). They also speak to substrate independence: the bench's actuators are simple pull-only

opponent motors, not the Hill-type muscle of the reference simulations, so the registers it reproduces already hold in a different physical realization. And the confirm-and-correct ledger is falsification in practice — the architecture's own predictions were put at risk, and three were corrected.

9.0.0.2 *The decisive measurement still owed.*

The architecture's strongest and riskiest claim is that alert energy is a *real metabolic quantity*. The bench can show its functional consequences (Pred-3), but cannot measure biological metabolism. The clean test is in the wet lab: combined EMG and metabolic measurement (NIRS, ³¹P-MRS) during sustained isometric tasks, looking for antagonist tonic activation that tracks agonist load linearly (register 1) and decays over seconds to tens of seconds after release rather than vanishing immediately. A positive result would distinguish the SMN from both classical reciprocal inhibition and the bare equilibrium-point hypothesis; a null would falsify its central energetic commitment. We name this as the experiment that matters most and that the bench cannot run.

9.0.0.3 *What a simulation can and cannot carry.*

A simulation is not a body and has no phenomenology; nothing here is claimed to feel anything. What a physics engine does carry is the structural and relational conditions the SMN holds to constitute object-directedness — a body-relative frame, the reafferent separation of self from world, the dual interface, and the halt at resistance. These are what the bench measures. Consistent with this restraint, the world estimate is deflationary: for the living agent there is no stored, contentful world-model, only a current attunement constituted in coupling (Section 4). Whether the structural signatures are sufficient for experience, or merely necessary, is the question the architecture sharpens, not one it settles.

9.0.0.4 *Forthcoming transitions.*

Two threads are deliberately left open. The negotiable-pattern layer (Section 5) carries the architecture toward generative structure; its full development — how negotiable and transactional action patterns ground generative grammar — is the subject of a companion paper (Nagarjuna and Karnam, 2026). And the passage from the body-grounded snapshot to externalized, serialized meaning — the grounded-to-positional transition that conceptual spaces mediate — is the subject of the second monograph in this program. The present paper establishes the basal and haltable layers both build on.

10 CONCLUSION

We have proposed an architecture in which cognition is the modulation of sensation through action, built from opponent pairs of elements that sense and act through one substrate, coordinated by a decentralized broadcast, and holding themselves in dynamic equilibrium against the fields they live in. Its organizing claim is that *haltability* — the active, metabolically costly hold an antagonistic body can recruit — is the architectural ground of object-directedness: an object is registered as *other* in the moment an unpredicted resistance halts an ongoing action. From this the architecture earns the self/world distinction (reaffERENCE), attention (surprise-driven halt), and a route, stated here as a hypothesis, from negotiable action patterns to generative structure. The account keeps what cognitivism and 4E each get right — generativity and embodiment — while supplying the mechanism each leaves open, and it commits to eight measurable registers and a confirm-and-correct computational ledger that put it at risk. The single measurement it still owes — the metabolic correlate of alert energy — is the experiment we most want done. A theory worth proposing says clearly what would refute it; this is ours.

CONFLICT OF INTEREST STATEMENT

341 The authors declare that the research was conducted in the absence of any commercial or financial
342 relationships that could be construed as a potential conflict of interest.

AUTHOR CONTRIBUTIONS

343 GN and DK jointly developed the SMN architecture and its theoretical argument, over a sustained
344 collaboration of about ten years carried out alongside their professional commitments as science-education
345 researchers. The biological and philosophical grounding, and the recognition of the need to seek an
346 alternative framework, originated with GN. SMN-Lab was developed by GN with AI-assisted code.

FUNDING

347 This work was not supported by any grant. The early research was carried out as part of the knowledge
348 lab at the Homi Bhabha Centre for Science Education (HBCSE), Tata Institute of Fundamental Research
349 (TIFR), under the project “The Roots of STEM,” until GN’s superannuation in April 2021.

DATA AVAILABILITY STATEMENT

350 All code, documentation, and seeded datasets for SMN-Lab are openly available at [https://github](https://github.com/gnowgi/smn-lab)
351 [.com/gnowgi/smn-lab](https://github.com/gnowgi/smn-lab) (GPL-3.0); documentation at <https://smn-lab.readthedocs.io>.

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